STAT 405 Final Project Proposal

1. Team Information

Team member:

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2. Research Question

This project applies Bayesian inference to baseball batting performance. We will use the 500hits dataset, which contains career batting statistics for 463 baseball players. In particular, we will focus on the variables AB (total at-bats) and H (total hits).

Our main scientific question is: **Which player has the highest underlying hitting probability after accounting for uncertainty, and how do player rankings change under different Bayesian models and posterior computation methods?**

3. Candidate Datasets

Two candidate datasets have been identified for this project.

Main dataset: Baseball Player Metrics

Our main dataset is the Baseball Player Metrics / 500hits dataset. It contains career-level statistics for 463 baseball players and includes variables such as years played, games, at-bats, runs, hits, doubles, triples, home runs, and RBIs.

We will primarily use:

- Player: player name
- YRS: number of years active in professional baseball
- G: total games played
- AB: total at-bats
- H: total hits
- BA: observed batting average

This dataset is appropriate for our project because for each player, the number of hits can naturally be modelled as a Binomial outcome with the number of trials equal to at-bats. This gives a clean and interpretable Bayesian modelling framework.

Here is the structure of the dataset:

```
##             PLAYER YRS    G    AB    R    H X2B X3B  HR  RBI   BB   SO  SB  CS
## 1          Ty Cobb  24 3035 11434 2246 4189 724 295 117  726 1249  357 892 178
## 2      Stan Musial  22 3026 10972 1949 3630 725 177 475 1951 1599  696  78  31
## 3     Tris Speaker  22 2789 10195 1882 3514 792 222 117  724 1381  220 432 129
## 4      Derek Jeter  20 2747 11195 1923 3465 544  66 260 1311 1082 1840 358  97
## 5     Honus Wagner  21 2792 10430 1736 3430 640 252 101    0  963  327 722  15
## 6 Carl Yastrzemski  23 3308 11988 1816 3419 646  59 452 1844 1845 1393 168 116
##      BA HOF
## 1 0.366   1
## 2 0.331   1
## 3 0.345   1
## 4 0.310   1
## 5 0.329   1
## 6 0.285   1
```

Source:

https://www.kaggle.com/datasets/krupadharamshi/500hits/data

Dataset 2 (Backup Dataset)

Dataset name: Gallstone Disease Dataset (UCI Machine Learning Repository)

Source:

https://archive.ics.uci.edu/dataset/1150/gallstone-1

This dataset contains clinical and demographic information from 319 individuals, including 161 patients diagnosed with gallstone disease. The dataset includes 38 features such as demographic information, bioimpedance measurements, and laboratory test results.

The outcome variable is:

Gallstone = 0 or 1

If necessary, this dataset is suitable as a backup dataset because it contains multiple clinical variables related to gallstone risk factors and can be analyzed using Bayesian logistic regression.

Here is the structure of dataframe:

	Gallstone.Status <int>	Age <int>	Gender <int>	Comorbidity <int>	Coronary.Artery.Disease..CAD. <int>	Hypothyroidism <int>	Hyperlipidemia <int>
1	0	50	0	0	0	0	0
2	0	47	0	1	0	0	0
3	0	61	0	0	0	0	0
4	0	41	0	0	0	0	0
5	0	42	0	0	0	0	0
6	0	96	0	0	0	0	0

6 rows | 1-8 of 39 columns

4. **Proposed Models**

We propose to analyze the data using two Bayesian models.

Model 1:

Our first model treats each player independently. It is a simpler Bayesian model where we assume

$$H_i | P_i \sim Binomial(AB_i, \ p_i)$$

with a prior on each player's hitting probability $p_i \sim Beta(\alpha, \beta)$.

In this model, each player has a separate parameter p_i; they are all independent. This is a simple baseline model. For each player, we will compute the posterior distribution of p_i, use the posterior mean as a point estimate, and use credible intervals to measure uncertainty. Then we will rank players by their posterior means.

Model 2:

Our second model is a hierarchical model. In this model, we do not treat each player completely separately. Instead, we assume that all players' true hitting probabilities come from one common population distribution.

For a player i, let H_i be the number of hits and AB_i be the number of at-bats, we assume $H_i | P_i \sim Binomial(AB_i, \ p_i)$, where p_i is the true hitting probability of the player i.

The difference here is that P_i they are now related to each other through $p_i \sim Beta(a, b)$, where a and b are unknown hyperparameters and also have priors.

This model allows information sharing across players, so it can shrink extreme estimates toward the overall average.

5. **Posterior Computation Methods**

The posterior distribution will be approximated using different methods; our goal is to compute posterior means and credible intervals under each model and compare the ranking results.

For simpler model 1, we can use SNIS or laplace approximation.

For hierarchical model 2, we can use HMC and Variational inference.

6. **Model Comparison and Evaluation**

We will compare both the two models and the posterior computation methods used in the project. First, we will compare the no-pooling model and the hierarchical model to study how partial pooling changes posterior means, credible intervals, and player rankings.

Second, we will compare the posterior results obtained from different computation methods. For the simpler model, we will compare SNIS and the Laplace approximation. For the hierarchical model, we will compare HMC and Variational Inference. In this way, we can separately study the effect of model choice and the effect of the posterior computation method.

7. **Implementation Plan**

The project will proceed in the following steps:

1. Data cleaning and exploratory data analysis
2. Implementation of two models
3. Posterior inference using SNIS and Laplace approximation
4. Posterior inference using HMC and VI
5. Model comparison and diagnostic analysis
6. Preparation of the final report

8. **Contribution Plan**

We will divide the work equally. One team member will focus on data cleaning, exploratory analysis, and implementation of the no-pooling model. The other team member will focus on the hierarchical model and posterior computation. We will do a model comparison together. Both team members will contribute to writing, interpretation, and final editing.

9. **Git Repository**

The code for this project will be maintained in a public Git repository.

Repository link: https://github.com/Cindywlj/stat405-final-project

The repository will include data preprocessing scripts, model implementation code, and scripts for generating figures and tables used in the report.